INCH–POUND

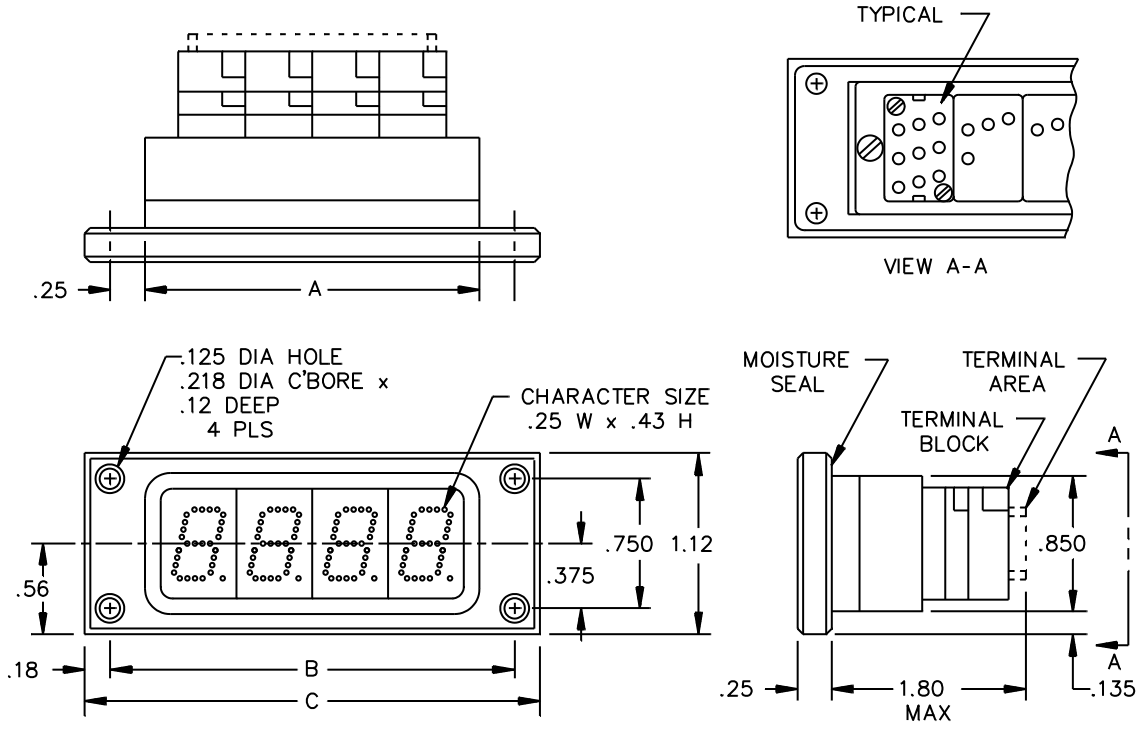
MIL–DTL–28803/2B
w/AMENDMENT 1
2 September 2018
SUPERSEDING
MIL–D–28803/2B
5 September 2013

DETAIL SPECIFICATION SHEET

# DISPLAY, OPTOELECTRONIC, SEGMENTED READOUT, BACKLIGHTED, STYLE I (INCANDESCENT LAMPS) RFI SHIELDED, MOISTURE SEALED, HIGH IMPACT SHOCK, TYPE R02

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–DTL–28803.




| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .12 | 3.0 | .18 | 4.6 | .375 | 9.52 | .750 | 19.05 |
| .125 | 3.18 | .218 | 5.54 | .43 | 10.9 | .850 | 21.59 |
| .135 | 3.43 | .25 | 6.4 | .56 | 14.2 | 1.12 | 28.5 |
| | | | | | | 1.80 | 45.7 |

FIGURE 1. Type R02 display.

AMSC N/A

FSC 5980

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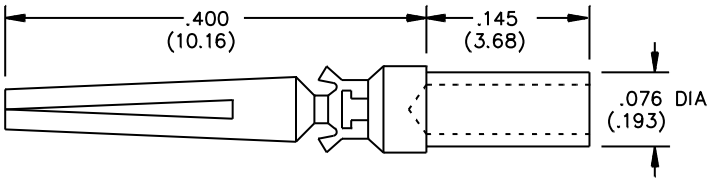
NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Design configuration optional within envelope as shown.
4. The values for dimensions A, B, and C are listed in table I.

FIGURE 1. Type R02 display – Continued.

TABLE I. Display assembly dimensions. 1/ 2/ 3/

| Number of units per assembly | A | B | C | Weight maximum |
|---|---|---|---|---|
| 3 | 2.00 (50.8) | 2.50 (63.5) | 2.87 (72.9) | 4.5 (127.6) |
| 4 | 2.50 (63.5) | 3.00 (76.2) | 3.27 (83.1) | 5.0 (141.7) |
| 5 | 3.00 (76.2) | 3.50 (88.9) | 3.87 (98.3) | 5.5 (155.9) |
| 6 | 3.50 (88.9) | 4.00 (101.6) | 4.37 (110.0) | 6.0 (170.0) |
| 7 | 4.00 (101.6) | 4.50 (114.3) | 4.87 (123.7) | 6.5 (184.3) |
| 8 | 4.50 (114.3) | 5.00 (127.0) | 5.37 (136.4) | 7.0 (198.4) |
| 9 | 5.00 (127.0) | 5.50 (139.7) | 5.87 (149.1) | 7.5 (212.6) |
| 10 | 5.50 (139.7) | 6.00 (152.4) | 6.87 (174.5) | 8.0 (226.8) |

1/ Dimensions are in inches. Millimeter equivalents are given for general information only.
2/ Weights are in ounces. Gram equivalents are given for general information only.
3/ Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.

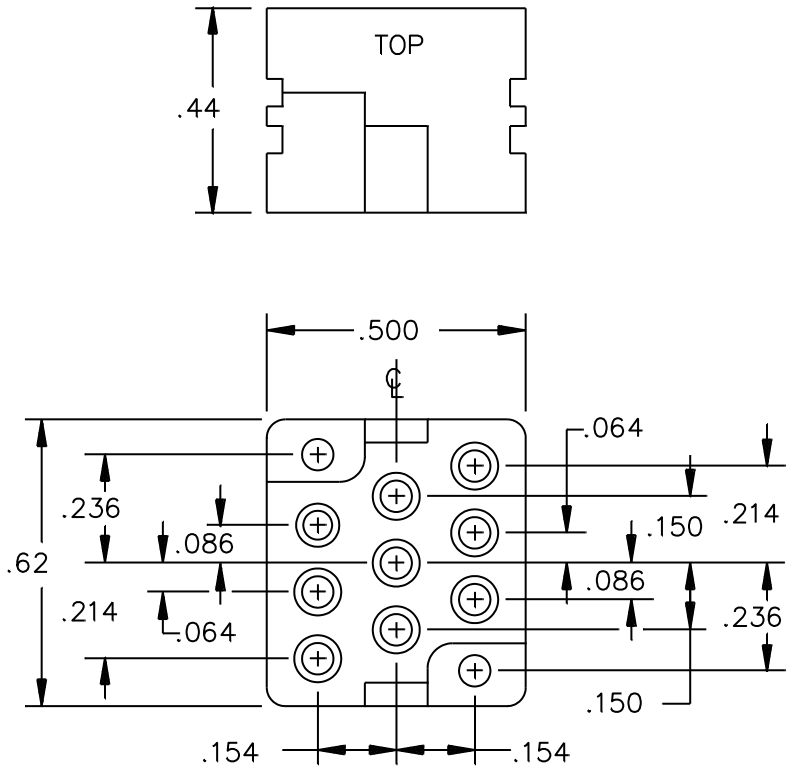



NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.010 inch (0.25 mm).
3. Design configuration optional within envelope as shown.
4. Terminal part number (package of 25 terminals): MIL–PRF–22885/80–50.
5. Application information:
   - Crimp tool: SAE-AS22520/1–01
   - Positioner: SAE-AS22520/1–02
   - Insertion tool: Safran Electronics & Defense (CAGE 96182) 800–3191
   - Extraction tool: Safran Electronics & Defense 800–P7
     Astro Tool Corp. (CAGE 58164) ATMS 2062
   - Locator tool: Safran Electronics & Defense 800–3191–L20
6. Terminals can be installed in terminal blocks without the use of special tools, but tool part number 800–P2 (CAGE 96182) is required for removal.

FIGURE 2. Terminal, crimp type.



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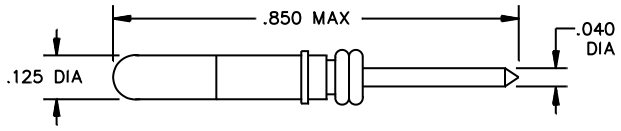

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .064 | 1.63 | .154 | 3.91 | .44 | 11.2 |
| .086 | 2.18 | .214 | 5.44 | .500 | 12.70 |
| .150 | 3.81 | .236 | 5.99 | .62 | 15.7 |

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Design configuration optional within envelope as shown.

FIGURE 3. Connector block.



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| Inches | mm |
|---|---|
| .040 | 1.02 |
| .125 | 3.18 |
| .850 | 21.59 |

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.010 inch (0.25 mm).
3. Design configuration optional within envelope as shown.

FIGURE 4. Lamp assembly.

TABLE II. Truth table for seven segment numeric alpha display (direct wired).

| Input | | | | | | | | Display | Output (Δ = Segment lit) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DPT | A | B | C | D | E | F | G | | A | B | C | D | E | F | G | DPT |
| | 0 | 0 | 0 | 0 | 0 | 0 | | 0 | Δ | Δ | Δ | Δ | Δ | Δ | | |
| | | 0 | 0 | | | | | 1 | | Δ | Δ | | | | | |
| | 0 | 0 | | 0 | 0 | | 0 | 2 | Δ | Δ | | Δ | Δ | | Δ | |
| | 0 | 0 | 0 | 0 | | | 0 | 3 | Δ | Δ | Δ | Δ | | | Δ | |
| | | 0 | 0 | | | 0 | 0 | 4 | | Δ | Δ | | | Δ | Δ | |
| | 0 | | 0 | 0 | | 0 | 0 | 5 | Δ | | Δ | Δ | | Δ | Δ | |
| | 0 | | 0 | 0 | 0 | 0 | 0 | 6 | Δ | | Δ | Δ | Δ | Δ | Δ | |
| | 0 | 0 | 0 | | | | | 7 | Δ | Δ | Δ | | | | | |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 8 | Δ | Δ | Δ | Δ | Δ | Δ | Δ | |
| | 0 | 0 | 0 | | | 0 | 0 | 9 | Δ | Δ | Δ | | | Δ | Δ | |
| | 0 | 0 | 0 | | 0 | 0 | 0 | A | Δ | Δ | Δ | | Δ | Δ | Δ | |
| | | | 0 | 0 | 0 | 0 | 0 | b | | | Δ | Δ | Δ | Δ | Δ | |
| | 0 | | | 0 | 0 | 0 | | C | Δ | | | Δ | Δ | Δ | | |
| | | 0 | 0 | 0 | 0 | | 0 | d | | Δ | Δ | Δ | Δ | | Δ | |
| | 0 | | | 0 | 0 | 0 | 0 | E | Δ | | | Δ | Δ | Δ | Δ | |
| | 0 | | | | 0 | 0 | 0 | F | Δ | | | | Δ | Δ | Δ | |
| 0 | | | | | | | | . | | | | | | | | Δ |



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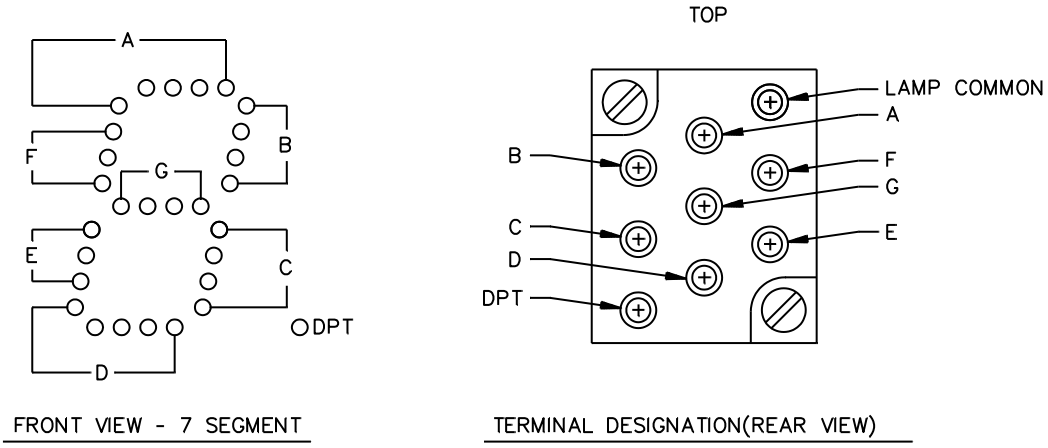



FIGURE 5. Truth table letter assignments for seven segment alpha-numeric display with decimal point.

TABLE III. Truth table for a four segment sign display with decimal point (direct wired).

| Input | | | | | Display | Δ = Segment lit | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| DPT | B | C | E | F | | B | C | E | F | DPT |
| | O | O | O | O | ⚡ | Δ | Δ | Δ | Δ | |
| | O | | O | | ····· | Δ | | Δ | | |
| O | | | | | • | | | | | Δ |

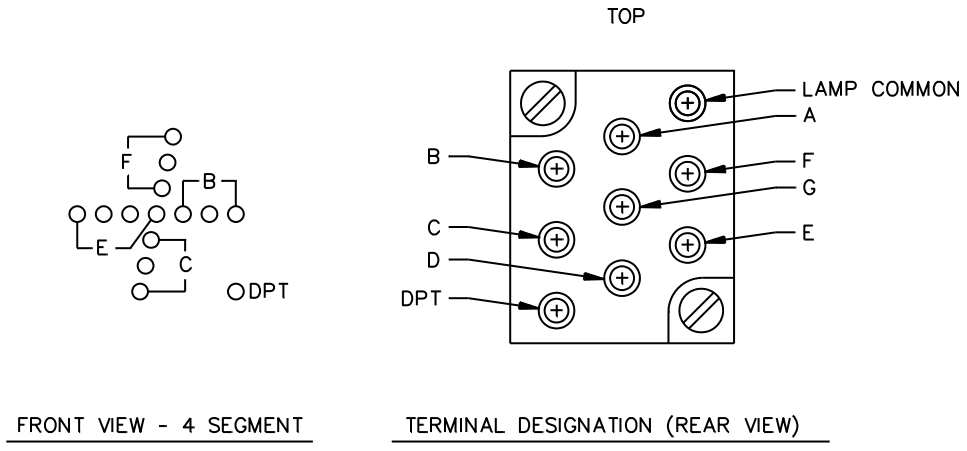



FIGURE 6. Truth table letter assignments for four segment sign display with decimal point.



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REQUIREMENTS:

Design and construction:

Dimensions and configurations. See figures 1 through 6 and tables I through V.

Display assembly. The display assembly shall be assembled and shall include the following:

a. Display modules.

b. Bezel assembly with lens, panel gasket, and fasteners.

c. Lamp assemblies (see figure 4).

d. Connector blocks (see figure 3).

e. Insertable crimp type terminals (packaged separately) (see figure 2).

Front lens plate: Lens material used to convert clear incandescent lamps to colored lights.

Light source: T–1 incandescent lamps.

Operating:

Dot to dot ratio: 1 to 1 maximum.

Code letters for module combinations and color and luminance: See figures 5 and 6 and tables IV and V.

Optical:

Chromaticity: Method I in accordance with MIL–DTL–28803.

Contrast ratio: C = 5 to 1 minimum.
$C^2$ = 4 to 1 minimum.

Physical and mechanical:

Mechanical endurance and solderability: Not applicable.

Terminal strength: The terminals shall be tested in accordance with MIL–STD–202–211, with the following details and exceptions: Test condition A and the applied force shall be 5 pounds (2.27 Kg).

Electrical:

Supply current: 5 V ±.25 V.

Supply voltage (for each lamp): .020 ampere ±.002 ampere.

Environmental:

Shock: Method II (high impact) in accordance with MIL–DTL–28803.

Vibration grade 2: Displays shall be tested in accordance with test condition A (10-500 Hz) of MIL–STD–202–204.



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TABLE IV. Code letters for number, type and arrangements of modules.

| Code letter | Number of modules | Type of module and arrangement |
|---|---|---|
| A | 3 module assembly | All 7 segment modules |
| B | 4 module assembly | All 7 segment modules |
| C | 5 module assembly | All 7 segment modules |
| D | 6 module assembly | All 7 segment modules |
| E | 7 module assembly | All 7 segment modules |
| F | 8 module assembly | All 7 segment modules |
| G | 9 module assembly | All 7 segment modules |
| H | 10 module assembly | All 7 segment modules |
| J | 3 module assembly | One 4 segment followed by two 7 segments |
| K | 4 module assembly | One 4 segment followed by three 7 segments |
| L | 5 module assembly | One 4 segment followed by four 7 segments |
| M | 6 module assembly | One 4 segment followed by five 7 segments |
| N | 7 module assembly | One 4 segment followed by six 7 segments |
| P | 8 module assembly | One 4 segment followed by seven 7 segments |
| Q | 9 module assembly | One 4 segment followed by eight 7 segments |
| R | 10 module assembly | One 4 segment followed by nine 7 segments |

TABLE V. Code letters for color and luminance combinations.

| Code letter | Color | 2100° Kelvin limits of chromaticity | | Luminance value in foot-lamberts (Candela/m²) minimum |
|---|---|---|---|---|
| | | X | Y | |
| A | White | 1/ | 1/ | 300 (1027.9) |
| B | Red | .695<br>.703<br>.655<br>.660 | .285<br>SL 2/<br>.325<br>SL 2/ | 50 (171.3) |
| C | Green | .260<br>.335<br>.018 | .570<br>SL 2/<br>SL 2/ | 25 (85.6) |
| D | Yellow | .562<br>.570<br>.596<br>.605 | .415<br>SL 2/<br>.382<br>SL 2/ | 200 (685.3) |

1/ Unfiltered, incandescent light.
2/ SL-Spectrum locus (where intersected by other coordinates pair).



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QUALIFICATION:

Extent of qualification: If suppliers are qualified to MIL–DTL–28803/1, table VI applies. If manufacturers are not qualified MIL–DTL–28803/1, then table VII applies.

TABLE VI. Extent of qualification for suppliers qualified to MIL–DTL–28803/1.

| PIN | Number of samples required | Tests of MIL–DTL–28803 | Extent of approval |
|---|---|---|---|
| M28803/2–RA | 2 | Group I | All |
| M28803/2–R 1/ | 10 | Group IV | |

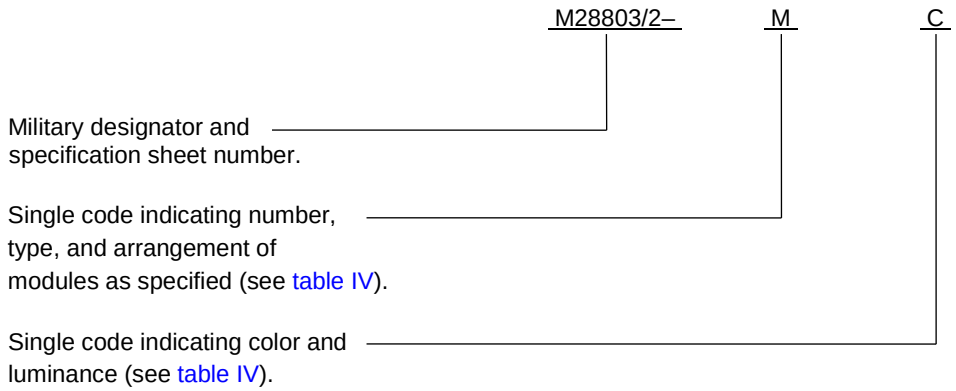
1/ Two sample units for each color.

TABLE VII. Extent of qualification for suppliers not qualified to MIL–DTL–28803/1.

| PIN | Number of samples required | Tests of MIL–DTL–28803 | Extent of approval |
|---|---|---|---|
| M28803/2–JA | 4 | Complete in accordance with qualification inspection of MIL–DTL–28803 | All |
| M28803/2–PB | 2 | | |
| M28803/2–A 1/ | 10 | | |

1/ Two sample units for each color.

Part or Identifying Numbers (PIN): The PIN shall consist of the letter "M" followed by the basic number of the specification sheet and a coded two-letter symbol indicating the number, type, and arrangement of the modules, and color and luminance, as specified. See example below:




PIN for terminals (see figure 2) (package of 25): MIL–PRF–22885/80–50.

PIN for connector block (see figure 3) and terminals (see figure 2, 11 each): M28803/2–WR.

PIN for lamp assembly (see figure 4): M28803/2–WS.



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Reference documents. In addition to MIL–DTL–28803, this document references the following:

MIL–STD–202–204
MIL–STD–202–211
MIL–PRF–22885/80
MIL–DTL–28803/1

SAE AS22520/1

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Navy – EC
Air Force – 85
DLA – CC

Review activities:
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Air Force – 71

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